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Full Length Article

Flash SewerNet: A lightweight expert network with Gaussian-multimodal fusion for sewer defect recognition

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ABSTRACT

Sewer defect recognition from video inspections is a critical but challenging task due to three persistent issues in such harsh environments: severe image contrast degradation from sensor noise, which obscures defect features; high cross-similarity between different defect and material classes; and the difficulty in distinguishing infrequent defect types from the background, given scarce samples and weak features. These challenges significantly limit the accuracy, robustness, and practical deployment of existing models. To overcome these limitations, we propose Flash SewerNet, a lightweight expert network that effectively incorporates multimodal prior knowledge for robust sewer defect recognition. This network synergizes Gaussian modeling with multimodal priors, paired with an expert-assisted multi-label recognition head. This architecture is specifically co-designed for high stability and computational efficiency, making it suitable for real-time video analysis on resource-limited edge devices. Extensive experiments show that Flash SewerNet achieves state-of-the-art performance on several sewer inspection benchmarks. More importantly, we demonstrate its practical viability by deploying the model on an embedded platform, where it achieves a high inference speed sufficient for continuous, in-situ defect diagnosis in real-world sewer inspection videos. This work not only presents a robust detection solution but also highlights the essential role of multimodal domain knowledge in advancing vision systems for challenging real-world applications. The code and models are available at: https://github.com/zkkzxk/Flash-SewerNet.

1. Introduction

As a crucial component of urban infrastructure, the operational condition of sewer systems (Chen et al., 2024; Hu et al., 2024) directly influences flood control, environmental safety, and public health. In recent years, vision-based intelligent detection technologies (Fan et al., 2024; Rayhana et al., 2023; Yang et al., 2023) have emerged as vital tools for identifying sewer defects, owing to their efficiency and non-invasive nature. A significant development in this domain has been the introduction of models capable of generating defect localization heatmaps alongside recognition results, offering valuable visual explanations for diagnostic decisions. Nevertheless, the effectiveness of such explainable systems hinges fundamentally on the performance of their underlying recognition algorithms. In real-world sewer environments, where multiple defect types frequently co-occur, the limitations of current recognition methods become particularly evident—compromising not only classification accuracy but also the reliability of defect localization. To address this challenge, this work focuses on advancing multi-label recognition techniques (Jin et al., 2025; Zhang et al., 2025), with the goal of establishing a more robust foundation for accurate defect identification and trustworthy visual interpretation, ultimately enhancing the efficacy of sewer maintenance.

The early methods mainly relied on traditional image processing techniques (Myrans et al., 2018a; Su & Yang, 2014) for multi-label defect recognition. Although these approaches worked well in limited settings, they depended heavily on manually designed features, were highly sensitive to noise, and struggled with complex situations involving overlapping defects. Additionally, variations in sewer material properties led to significant differences in defect textures, highlighting the poor generalizability of traditional methods. These limitations collectively rendered traditional approaches inadequate for robust, large-scale deployment in diverse sewer environments.

In recent years, deep learning has shown great promise in multi-label image recognition tasks due to its strong feature learning capabilities (Dian et al., 2022; Guan et al., 2026; Yan et al., 2024; Zhao et al., 2023). Unlike traditional techniques, deep neural networks can automatically extract multi-scale, high-level visual features and effectively capture relationships among complex defects, providing innovative solutions for multi-label defect recognition in sewer systems. Notable advances have been made along several fronts. KSSNet (Wang et al., 2020)

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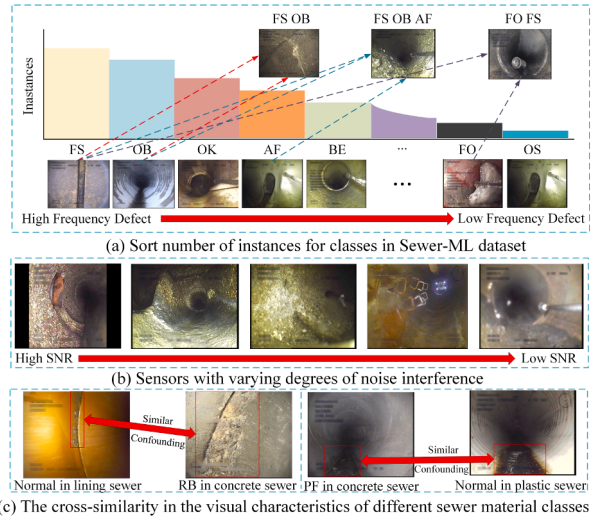


Fig. 1. The figure illustrates three key challenges in multi-label sewer defect recognition:(a) Class imbalance: A bar chart reveals a long-tail distribution, with high-frequency defects abundant and low-frequency ones scarce, hindering model generalization.(b) Sensor noise: High-SNR sensors capture clear features, while low-SNR ones in sewers suffer noise, blurring textures and obscuring defects.(c) Cross-material similarity: Visual confounding exists across sewer materials, increasing recognition ambiguity.

leverages Graph Convolutional Networks to model semantic correlations among defect labels, while Liu et al. (2024b) propose a cross-modal alignment framework that enriches visual features using vision-language pre-trained models. Another line of work focuses on enhancing feature robustness and global perception: Hu et al. (2023) introduce a self-purification mechanism to combat noise-induced degradation, and Haurum et al. (2022a) adopt multi-scale hybrid Transformers for comprehensive global context modeling. Separately, Zhong et al. (2025) tackle the long-tailed distribution problem via label-guided knowledge distillation.

However, a critical survey of the literature shows that no single deep learning method to date has been able to adequately address three inherent challenges that are singular to real-world sewer inspection, as illustrated in Fig. 1: (1) pervasive sensor noise and low image quality that degrade feature reliability; (2) high visual similarity across different materials and defect types that complicates accurate recognition; and (3) a long-tailed distribution of defect classes that causes rare but critical defects to be overlooked. These factors collectively undermine the robustness of automated inspection systems.

Firstly, factors such as uneven lighting, moisture scattering, and limited sensor dynamic range in confined spaces greatly reduce image quality. When deep neural networks analyze these images, the gradient features of defects like cracks become heavily weakened, making reliable defect recognition difficult, particularly when features are blurred or incomplete. Secondly, defects occurring in various sewer materials may have different physical origins but display very similar visual patterns, complicating accurate recognition based solely on image data. For instance, a normal lined sewer with attached debris and a cracked concrete sewer can exhibit comparable spatial frequency distributions in their texture features. Thirdly, sewer defect recognition suffers from a pronounced long-tail distribution problem, where the ratio of samples between common and rare defect classes can be as high as 100 to 1. During training, the gradient signals from rare classes are overshadowed by those from common classes, leading the model to often misclassify ambiguous features from rare classes as background.

To address three major challenges in multi-label sewer defect recognition, this study proposes Flash SewerNet, a lightweight expert network that synergistically integrates Gaussian modeling with multimodal prior knowledge. Unlike existing frameworks that rely on deterministic feature extractors and single-modal supervision, our approach is distinguished by three core innovations that fundamentally depart from current multimodal and mixture-of-experts architectures. First, we replace point-based features with a Gaussian distribution-based spatial encoding that explicitly models feature uncertainty, significantly enhancing robustness against sensor noise and illumination variations—a critical advantage in harsh sewer environments. Second, we introduce sewer material as a high-level semantic prior to dynamically guide spatial attention, enabling the network to adapt feature extraction to different pipe substrates—a capability absent in current uniform-treatment methods. Unlike standard multimodal vision-language approaches (Wang et al., 2020; Yan et al., 2024) that simply concatenate visual and textual features with heavy encoders at inference, our multimodal fusion performs deep semantic guidance where text embeddings actively reweight visual channels during feature extraction, with the text encoder used only during training. On the decision side, we depart from conventional single-stream classifiers and generic mixture-of-experts (MoE) architectures by designing a mixture-of-experts with assistants (MoEA) system that mimics a judicial deliberation process: the assistant module captures locally salient cues while the expert module performs global contextual reasoning, with a dual-constraint mechanism ensuring both decision balance and diversity—explicitly engineered to counteract the long-tailed distribution problem that generic MoE overlooks. This dual-path collaborative architecture, combined with semantically guided multimodal fusion, allows Flash SewerNet to adaptively optimize feature representations according to semantic context—markedly improving discrimination of rare defect classes and robustness against complex backgrounds. Collectively, these innovations establish a new paradigm that unites traditional image processing principles with deep learning, moving beyond the limitations of existing systems that treat feature extraction and decision-making as isolated, deterministic sewer.

To comprehensively evaluate the performance of Flash SewerNet, we conducted systematic experimental validation on three benchmark datasets: Sewer-ML (Haurum & Moeslund, 2021), a specialized urban drainage dataset from Chengdu, China, and the challenging QV-Pipe (Liu et al., 2022a) video benchmark. The results demonstrate that our model achieves SOTA performance on Sewer-ML dataset while maintaining its lightweight design. Furthermore, experiments on the Chengdu dataset confirm that sufficient multimodal textual information can provide richer environmental priors for the model, effectively enhancing its discriminative capability for visually ambiguous features. Additional evaluation on QV-Pipe further validates the model's effectiveness in real-world video inspection scenarios. The main contributions of this study are as follows.

1. We propose Flash SewerNet, a lightweight expert network for real-time sewer defect recognition. By integrating two core modules—a Multimodal Prior Feature Extraction (GMMPFE) network and a MoEA decision system—it offers a novel and efficient solution to three key challenges in multi-label defect recognition of sewer images.
2. Unlike conventional deterministic methods that are vulnerable to sensor degradation, we introduce a multi-scale Gaussian modeling mechanism that explicitly quantifies and mitigates feature uncertainty. This provides a robust low-level representation that preserves discriminative cues even under severe noise and poor illumination—fundamentally rethinking how sewer images should be represented.
3. Going beyond pure visual appearance, we propose a material-aware attention module that leverages sewer material text as a high-level semantic prior. This dynamically reweights spatial features to resolve cross-material visual similarity—a task-specific prior that existing methods overlook, yet is critical for reducing false positives in ambiguous scenarios.

4. To counteract the severe class imbalance in real-world sewer data, we design a dual-path MoEA architecture. Unlike standard Mixture-of-Experts that merely scale capacity, our assistant branch explicitly captures subtle local patterns of rare defects, while the expert branch performs global reasoning, jointly achieving balanced and accurate multi-label predictions through a novel dual-constraint optimization.

2. Related work

2.1. General multi-label recognition method

Multi-label image recognition involves predicting multiple object, scene, or attribute labels present simultaneously within an image. This task is notably more challenging than single-label recognition due to variations in the scale, location, and visibility of different targets, as well as the complex semantic and spatial relationships among labels. In recent years, deep learning techniques—particularly Convolutional Neural Networks (CNNs) and their variants—have become the dominant approach for multi-label recognition (Kuang & Li, 2026; Li et al., 2024; Ma et al., 2025; Sun et al., 2024), substantially enhancing model representational power and recognition accuracy. Early methods largely relied on handcrafted features and traditional machine learning algorithms, which often failed to capture intricate structures and nonlinear interdependencies among labels. With the advancement of deep learning, especially the success of CNNs on large-scale datasets such as ImageNet (Deng et al., 2009), researchers began adopting these models for multi-label tasks. CNNs automatically learn hierarchical image features through layered convolution and pooling operations, enabling powerful feature representations ranging from low-level edges and textures to high-level semantic concepts.

To better leverage semantic information from different image regions, attention mechanisms (Fu et al., 2025; Ge et al., 2021; Hu et al., 2018; Li et al., 2019) have been incorporated into multi-label frameworks. These modules dynamically focus on regions relevant to specific labels, thereby strengthening the model's ability to extract discriminative features. Studies have employed spatial or channel attention mechanisms to emphasize visual patterns associated with target labels through feature map weighting. This not only improves recognition accuracy but also enhances model interpretability by making the decision-making process more transparent.

Despite CNNs' strengths in feature extraction, their limited receptive field restricts their capacity to model global contextual information. To capture interdependencies between labels, some methods (Gürsoy & Kaya, 2024; Ye et al., 2020) integrate GCNs or Recurrent Neural Networks (RNNs). GCNs treat labels as graph nodes and model label correlations through message passing, incorporating semantic prior knowledge to boost recognition. For example, several works construct label co-occurrence graphs and apply GCNs to propagate semantic information and enrich visual representations. RNNs, on the other hand, predict labels sequentially and convey dependencies through hidden states. However, these approaches often require additional label relationship annotations or complex network designs, increasing model complexity.

Recently, the success of the Transformer (Vaswani et al., 2017) architecture in natural language processing has inspired its adoption in computer vision. Vision Transformer (ViT) (Liu et al., 2021) and related variants (Liu et al., 2024a, 2022b; Wang et al., 2021) use self-attention to model global relationships among image patches, effectively capturing long-range dependencies. In multi-label recognition, Transformers implicitly model label interactions via self-attention without requiring explicit prior knowledge of label relationships. Some studies divide images into sequences of patches, employ Transformer encoders to extract global features, and predict multiple labels through specialized heads. Other hybrid approaches combine the local feature extraction capabilities of CNNs with the global contextual model.

2.2. Multi-label sewer defect recognition method

In the development of multi-label sewer defect recognition technology, early research primarily relied on traditional image processing methods (Myrans et al., 2018b; Piciarelli et al., 2018), which employed handcrafted feature extraction algorithms for defect recognition. For example, Sinha and Fieguth (2006) proposed a neuro-fuzzy classifier combining neural networks and fuzzy logic to classify features extracted from segmented sewer images. Another method introduced morphological segmentation based on edge detection, including open-top-hat and close-bottom-hat transformations, for recognizing sewer defects. However, such approaches exhibited clear limitations when dealing with the complex and variable imaging conditions inside sewers. Their recognition accuracy and robustness heavily depended on the quality of feature engineering, making it difficult to adapt to the diversity of defect morphologies and complex background interference, thus limiting their generalizability.

With the rapid advancement of computer vision, deep learning-based methods have emerged as a revolutionary alternative in the field of automated sewer defect recognition. In the domain of CNN-based multi-label sewer defect recognition, researchers such as Yu et al. (2024), Xie et al. (2019), Hassan et al. (2019), and Myrans et al. (2019) have conducted a series of studies with notable results. Nevertheless, due to the significant commercial value of this field, most of these methods are based on different private datasets that vary considerably in defect classes and distributions. To enable fair and open comparison of multi-label recognition methods, Haurum and Moeslund (2021) introduced a new publicly available multi-label sewer defect dataset along with a novel evaluation metric, $F2_{CIW}$, tailored for sewer image defect recognition. Since then, numerous advanced methods have continued to emerge, significantly improving the accuracy of multi-label sewer recognition and pushing the technology toward practical applicability.

Notable innovations include: a self-purification module introduced by Hu et al. (2023), which effectively mitigates feature degradation caused by underwater imaging conditions through an adaptive feature purification mechanism, enhancing model robustness to noise; a multi-scale hybrid vision transformer proposed by Haurum et al. (2022a), which integrates the local feature extraction capabilities of CNNs with the global modeling ability of Transformers to accurately capture and fuse multi-scale sewer defect features; and a label-guided and tail-aware knowledge distillation strategy developed by Zhong et al. (2025), which leverages semantic relationships among labels to guide feature learning and uses knowledge distillation to alleviate recognition bias caused by scarce tail-class defect samples in long-tailed data distributions.

As shown in Table 1, existing multi-label sewer defect recognition methods can be broadly compared along three dimensions: robustness to sensor-induced feature degradation, exploitation of multimodal priors, and handling of long-tailed class distributions. Methods such as KSSNet (Wang et al., 2020) focus on modeling semantic correlations among defect labels but do not explicitly address feature reliability under noise. The self-purification method of Hu et al. (2023) targets noise-induced feature degradation, yet it relies solely on visual information and does not exploit material-level priors. The multi-scale hybrid Transformer of Haurum et al. (2022a) strengthens global context modeling but treats all sewer materials uniformly. The label-guided knowledge distillation of Zhong et al. (2025) addresses the long-tailed distribution through semantic label relationships, while CLIP-based approaches such as Yan-CLIP (Yan et al., 2024) introduce vision-language pretraining but still rely on heavy encoders at inference. In contrast, Flash SewerNet jointly addresses the three challenges within a single lightweight framework: the EA-Stem enhances contour features under degraded imaging, the GMMPFE module injects sewer material as a semantic prior through CLIP embeddings while keeping the text encoder out of inference, and the MoEA decision system counteracts the long-tailed distribution through expert specialization and score-weighted assistant

Table 1
Comparison of representative multi-label sewer defect recognition methods.

Method	Noise Handling	Multimodal Prior	Long-tail Strategy
KSSNet (Wang et al., 2020)	–	–	Label graph
(Hu et al., 2023)	Self-purification	–	–
(Haurum et al., 2022a)	–	–	–
(Zhong et al., 2025)	–	Label-guided	Knowledge distillation
Yan-CLIP (Yan et al., 2024)	–	CLIP text	–
Flash SewerNet	Gaussian smoothing	Material-aware CLIP prior	MoEA

selection. This combination distinguishes our method from prior work, which typically addresses only one or two of these aspects.

2.3. Multimodal prior knowledge

Multimodal prior knowledge guidance, as a cutting-edge research direction in the field of artificial intelligence, significantly enhances the perception and cognitive abilities of models in complex scenarios by integrating domain knowledge from different information sources. This knowledge integration method (Wiczling, 2025; Xie et al., 2025) can overcome the limitations of purely data-driven methods, making the model more generalized, interpretable, and practical. However, how to efficiently and accurately utilize multimodal prior knowledge and effectively integrate it into the model training and inference process to fully leverage its advantages remains a key challenge in current research. Especially in the task of multi label recognition of sewer image defects, this problem is particularly prominent. Sewer image data has complexity and specificity, with diverse types and shapes of defects, and is affected by various factors such as lighting, shooting angle, and dirt obstruction, which makes it difficult to extract image features and ensure recognition accuracy. At the same time, in addition to the image data itself, there is also a large amount of text information related to sewers, such as sewer materials, laying areas, etc. These text information contain rich prior knowledge and have important guiding significance for accurately identifying defect types in images. However, currently, there is insufficient research and effective solutions on how to deeply integrate these prior knowledge of text with image data, guide models to extract key features more accurately, and thereby improve the accuracy of multi-label recognition of sewer image defects. Based on this, this article mainly proposes an attention guided feature extraction method driven by prior knowledge using textual information such as sewer materials to assist in multi-label recognition of sewer image defects. This method aims to fully explore the intrinsic relationship between prior knowledge of text and image data, and dynamically adjust the model's attention to image features through attention mechanisms, so that the model can focus on the image regions most relevant to defect recognition, thereby improving the accuracy and reliability of recognition.

3. Method

To address the three core challenges of multi-label sewer defect recognition (sensor noise degrading feature reliability, cross-material visual similarity, and long-tailed class distribution), we propose Flash SewerNet, a lightweight expert network with Gaussian-multimodal fusion. The network adopts a multi-stage structural design, systematically and progressively extracting multi-scale features. Specifically, for stages 1 to 4, it uses downsampling rates of 1/4, 1/8, 1/16, and 1/32, respectively, to capture features at different granularities in different stages. This architecture consists of three main components: EA-Stem (Edge Aware Spatial Transformer Embedding Module), GMMPFE block and MoEA decision system, as shown in Fig. 2.

Table 2
Architecture configurations of Flash SewerNet.

Stage	Down. Rate	Layer	Input/Output channels Tiny	Small
1	$\frac{H}{4} \times \frac{W}{4}$	EA-Stem GMMPFE$\times N_1$	3/32 32/32	3/64 64/64
2	$\frac{H}{8} \times \frac{W}{8}$	Downsampling GMMPFE$\times N_2$	32/64 64/64	64/128 128/128
3	$\frac{H}{16} \times \frac{W}{16}$	Downsampling GMMPFE$\times N_3$	64/128 128/128	128/256 256/256
4	$\frac{H}{32} \times \frac{W}{32}$	Downsampling GMMPFE$\times N_4$	128/256 256/256	256/512 512/512
Number of Block			$[N_1, N_2, N_3, N_4] = [1, 4, 4, 2]$	

Among these components, the EA-Stem serves as the core module in the initial stage of the network. It deeply integrates Laplacian of Gaussian (LoG) filters with a CNN architecture to construct robust low-level feature representations with enhanced contour perception capabilities. Building upon this foundation, the GMMPFE module further suppresses sensor noise interference and mitigates visual feature ambiguity through Gaussian-based smoothing and multimodal feature injection. Finally, the MoEA decision system optimizes feature representation under long-tailed data distribution, forming a multi-label sewer defect recognition network that achieves both high accuracy and computational efficiency. The detailed architecture of the Flash SewerNet model is summarized in Table 2.

3.1. EA-stem layer

To address the challenge of visual feature degradation in enclosed sewer environments, we propose the EA-Stem module, which integrates Gaussian smoothing and the Laplacian operator to achieve simultaneous noise suppression and edge enhancement. The Gaussian kernel effectively mitigates image sensor noise, while the Laplacian operator

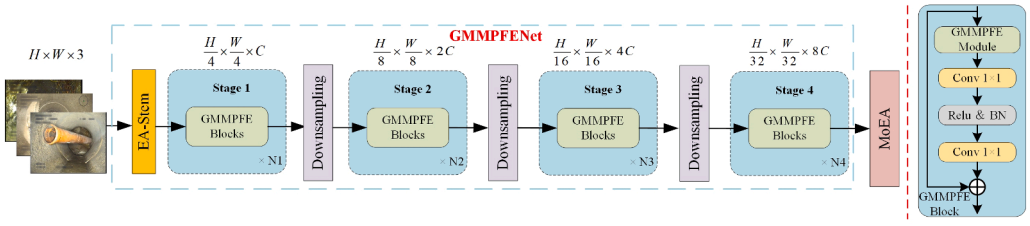


Fig. 2. Overview of Flash SewerNet architecture. It encompasses four stages. H, W and C denote the feature height, width, and channel width, respectively. The downsampling layer is composed of conventional CNNs.

strengthens contour responses. Even under complex and degraded imaging conditions, this module is able to extract robust low-level feature representations, providing a reliable foundation for subsequent sewer defect recognition tasks.

In our implementation, the input image $I \in \mathbb{R}^{H\times W\times 3}$ is first processed by a 7×7 convolutional layer to obtain an initial feature representation. A LoG filter with a 7×7 kernel and $\sigma = 1.0$ is applied to these features, explicitly enhancing the edge structures within the learned representation.

The 2D LoG filter is defined as:

$$\mathrm{LoG}_{\sigma}^{k\times k}(x,y) = \frac{1}{\pi\sigma^4}\left(1-\frac{x^2+y^2}{2\sigma^2}\right)e^{-(x^2+y^2)/(2\sigma^2)} \tag{1}$$

and the Gaussian filter as:

$$\mathrm{G}_{\sigma}^{k\times k}(x,y) = \frac{1}{2\pi\sigma^2}e^{-(x^2+y^2)/(2\sigma^2)} \tag{2}$$

To enhance feature representation and facilitate gradient flow, a residual connection is employed: the output of the LoG filter is first processed by activation function and then added to the original convolutional features.

$$\mathrm{F}_{LoG} = \mathrm{BN}\big(\mathrm{conv}_{7\times7}(I) + \mathrm{Sig}\big(\mathrm{LoG}_{\sigma}^{k\times k}\big(\mathrm{conv}_{7\times7}(I)\big)\big)\big) \tag{3}$$

where Sig denotes the Sigmoid activation function and BN denotes batch normalization.

Subsequently, two consecutive standard 3×3 convolutional operations are applied for downsampling, yielding feature map F_1.

$$F_1 = \mathrm{Conv}_{3\times3}\big(\mathrm{conv}_{3\times3}(F_{\mathrm{LOG}})\big) \tag{4}$$

This step reduces spatial dimensions to $\frac{H}{2}\times\frac{W}{2}$, facilitating efficient feature extraction.

To further refine the feature representation and incorporate multi-scale uncertainty information, F_1 is processed through two Gaussian filters with varying kernel sizes (7×7 and 5×5), ultimately achieving a final output resolution reduced to one-quarter of the original input.

$$F_{\text{EA-Stem}} = \big(G_{\sigma}^{7\times7}\big(\mathrm{Norm}\big(G_{\sigma}^{5\times5}(F_1)+F_1\big)\big)\big) \tag{5}$$

The integration of multi-scale Gaussian filters facilitates the combination of uncertainty-aware features across varying receptive fields, thereby enhancing the network's ability to capture and combine sewer image features at different resolutions.

3.2. GMMPFE block

To address the critical issues of sensor noise-induced low contrast and the cross-similarity in visual characteristics between different sewer materials and defect types, we introduce a feature extraction module grounded in a cooperative mechanism of Gaussian modeling and multimodal prior knowledge injection (e.g., sewer material). The proposed module, GMMPFE, is depicted in Fig. 3.

The feature extraction process of this module consists of three stages. In the first stage, textual descriptions T are fed into the CLIP model to generate embedded vectors F. These vectors are then mapped to the image feature space through a convolutional layer, yielding text-guided features F_T.

$$F_T = \mathrm{Conv}_{1\times1}(\mathrm{CLIP}(T)) \tag{6}$$

In the second stage, Gaussian modeling is applied to the input image features FIN to enhance the feature representation. This is followed by a residual connection and a convolutional layer to map the features, resulting in the enhanced image features F_I, thereby ensuring stable network training.

$$F_I = \mathrm{Conv}_{1\times1}\left(FIN + G_{\sigma}^{5\times5}(FIN)\right) \tag{7}$$

$$\sigma = \begin{cases} 0.2, & \text{Stage} = 1 \\ 1.0, & \text{otherwise} \end{cases} \tag{8}$$

It is noteworthy that as the network deepens, the standard deviation is set to 0.2 in Stage 1 and 1.0 in the remaining stages. This design is motivated by the fact that shallow layers require the preservation of detailed features, and hence a small standard deviation is adopted for precise filtering. In contrast, deeper layers need to integrate global semantic information, and a larger standard deviation facilitates cross-regional contextual fusion.

The third stage draws inspiration from the channel attention mechanism (Hu et al., 2018) and introduces textual information for multimodal guidance. Specifically, the image features F_I undergo Global Average Pooling (GAP) to compress spatial information. The compressed features are then subjected to matrix multiplication with the sewer material text features F_T, followed by a Sigmoid activation function to generate the channel weight vector. This process can be formally expressed as:

$$\mathrm{Weight} = Sig\big(\mathrm{GAP}\left(\mathbf{F}_I\right)\cdot\mathbf{F}_T\big) \tag{9}$$

The final enhanced features are combined with the input image features **I** through a residual connection. This operation not only preserves the original information but also achieves text-aware feature selection and fusion, enabling the model to adapt to visual feature variations under different sewer material conditions. Mathematically, this process can be expressed as:

$$F_{\text{enhanced}} = \mathrm{Weight}\cdot F_I + I \tag{10}$$

3.3. MoEA decision system

Compared to traditional fully connected layers, the MoE system offers significant advantages in multi-label sewer image recognition. It flexibly expands model capacity through sparse activation, enabling multiple expert networks to capture diverse defect patterns and environmental features without substantially increasing computational cost. This approach effectively models complex interdependencies among multiple labels and enhances model robustness through an ensemble learning mechanism. To further address the long-tail data distribution problem in sewer defect images, we introduce an assistant system within the MoE framework, which specializes in extracting subtle local features often overlooked in tail-class defect classes. This mechanism significantly improves recognition sensitivity to rare samples, mitigates

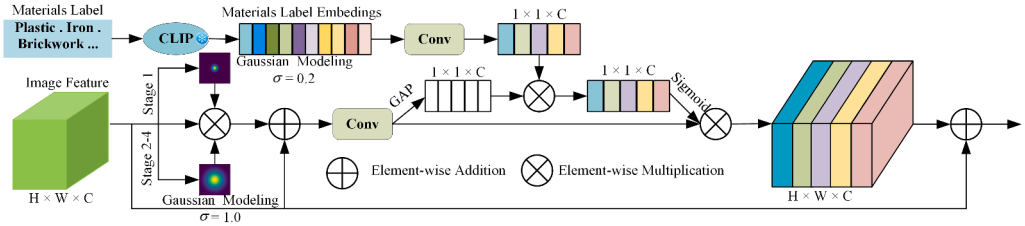


Fig. 3. Architecture of the GMMPFE block. The module consists of three stages: (1) text-guided feature generation using CLIP embeddings, (2) multi-scale Gaussian modeling for feature enhancement with adaptive standard deviation, and (3) multimodal fusion through channel attention mechanism that integrates textual prior knowledge with visual features.

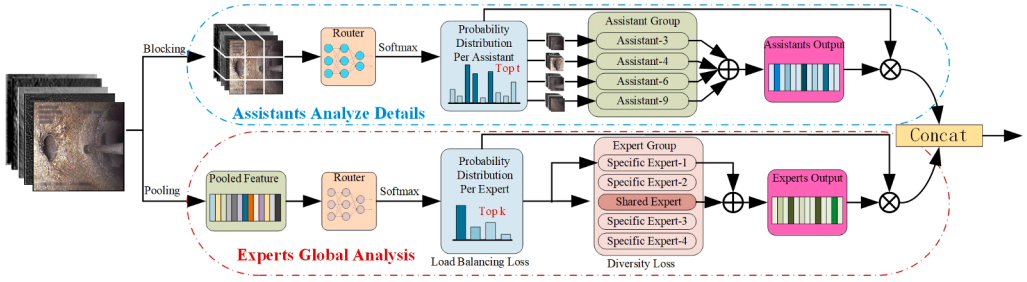


Fig. 4. Illustration of MoEA decision system. The system consists of two parallel branches: the MoE branch processes global features through expert networks with sparse activation, while the MoA branch performs spatial selection and enhancement to capture subtle local features.

recognition bias, and achieves more balanced and accurate multi-label predictions.

As illustrated in Fig. 4, the MoEA decision system comprises two distinct branches: a MoE system and a MoA system. The MoE branch processes the input feature tensor $\mathbf{X} \in \mathbb{R}^{B\times C\times H\times W}$ (where B, C, H, W denote batch size, channels, height, and width) by applying GAP to its spatial dimensions, transforming it into a vector representation:

$$x = \frac{1}{H\times W}\sum_{i=1}^{H}\sum_{j=1}^{W}\mathbf{X}_{:,:,i,j} \in \mathbb{R}^{B\times C} \tag{11}$$

The gating network generates selection weights for each expert based on the compressed feature vector:

$$\Pi = \text{softmax}(W_2 \cdot \text{GELU}(W_1 \cdot x) \in \mathbb{R}^{B\times E}) \tag{12}$$

where $W_1 \in \mathbb{R}^{C\times 2C}$ and $W_2 \in \mathbb{R}^{2C\times E}$ are parameter matrices of the gating network, E is the number of experts and GELU is Gaussian Error Linear Unit activation function.

Subsequently, the top k experts are selected for each sample and their weights are normalized:

$$\Pi_{\text{topk}}, \Gamma_{\text{topk}} = \text{top}_k(\Pi, k) \tag{13}$$

$$\hat{\Pi}_{\text{topk}} = \frac{\Pi_{\text{topk}}}{\sum_{j=1}^{k}\Pi_{\text{topk}}^{(j)}} \in \mathbb{R}^{B\times k} \tag{14}$$

The expert outputs are computed in a sparse manner, activating only the selected experts:

$$Y = \mathbf{0} \in \mathbb{R}^{B\times D} \tag{15}$$

$$\forall e \in [1, E]: \quad M_e = \left\{ i \in [1, B] \mid e \in \Gamma_{\text{topk}}^{(i)} \right\} \tag{16}$$

$$Y[M_e] = Y[M_e] + \hat{\Pi}_{\text{topk}}^{(e)} \odot f_e(X[M_e]) \tag{17}$$

where D is the output feature dimension, f_e denotes the forward propagation function of the eth expert and $\odot$ is Hadamard product.

The expert network is defined as:

$$f_e(X) = \text{Flatten}\big(W_{e,2} \cdot \text{GELU}\big(W_{e,1} \cdot X\big)\big) \tag{18}$$

where $W_{e,1} \in \mathbb{R}^{C\times h}$ and $W_{e,2} \in \mathbb{R}^{h\times D_{\text{out}}}$ are the parameter matrices of the eth expert, and h is the hidden dimension.

Additionally, we introduce two auxiliary losses to enhance training stability and expert diversity.

Load Balancing Loss (L_{bal}): Encourages the gating network to utilize all experts equally.

$$\mathcal{L}_{\text{bal}} = \sum_{e=1}^{E} p_e \log(p_e + \epsilon) + \log(E), \tag{19}$$

$$p_e = \frac{1}{B}\sum_{i=1}^{B}\Pi_{i,e} \tag{20}$$

where, $\epsilon > 0$ is a small constant added for numerical stability to prevent undefined behavior when $p_e = 0$.

Diversity Loss (L_{div}): Reduces similarity between expert outputs to avoid mode collapse.

$$\mathcal{L}_{\text{div}} = \frac{1}{|\mathcal{E}_{\text{active}}|^2}\sum_{e\neq e'}\text{ReLU}\left(\frac{f_e(x_e)^T f_{e'}(x_{e'})}{\|f_e(x_e)\|\|f_{e'}(x_{e'})\|}\right) \tag{21}$$

where $\mathcal{E}_{\text{active}}$ denotes the set of experts activated by the routing mechanism for the current batch, $f_e(x_e)$ is the output vector of expert e for its assigned inputs x_e.

The total auxiliary loss is the weighted sum of both:

$$\mathcal{L}_{\text{aux}} = \mathcal{L}_{\text{bal}} + \mathcal{L}_{\text{div}}, \tag{22}$$

Finally, the layer returns the aggregated expert output $Y \in \mathbb{R}^{B\times D_{\text{out}}}$ and the auxiliary loss $\mathcal{L}_{\text{aux}}$.

Meanwhile, the MoA system performs spatial dimension unfolding and gating computation on the input feature tensor X:

$$X_{\text{flat}} = X.\text{permute}(0, 2, 3, 1).\text{reshape}(B\times HW, C) \tag{23}$$

$$s = \text{Sig}\big(W_2 \cdot \text{ReLU}\big(W_1 \cdot X_{\text{flat}}\big)\big) \in \mathbb{R}^{B\times HW} \tag{24}$$

where $W_1 \in \mathbb{R}^{C\times(C/4)}$ and $W_2 \in \mathbb{R}^{(C/4)\times 1}$ are weight matrices of the gating network and s represents the activation scores for each spatial position.

Subsequently, the top t spatial positions with the highest scores are selected, where $t = \frac{H\times W}{4}$, and a score-weighted mask is constructed by retaining and normalizing the selected scores:

$$s_{\text{top}}, I_{\text{top}} = \text{top}_t(s, t), \tag{25}$$

$$\tilde{M} = \text{scatter}(0, I_{\text{top}}, 1) \odot s, \qquad M = \frac{\tilde{M}}{\sum_{j=1}^{HW}\tilde{M}^{(j)} + \epsilon} \in \mathbb{R}^{B\times HW} \tag{26}$$

where $\tilde{M}$ keeps the original gating scores at the selected positions, M is the normalized score-weighted mask, and ϵ is a small constant for numerical stability.

A shared transformation is applied to all positions to obtain enhanced feature representations:

$$H = \text{GELU}\big(X_{\text{flat}} \cdot W_{\text{shared}}\big) \in \mathbb{R}^{(B\times HW)\times 2C} \tag{27}$$

where $W_{\text{shared}} \in \mathbb{R}^{C\times 2C}$ is the weight matrix of the shared transformation layer.

Feature transformation is performed through position-specific auxiliary heads, and outputs from corresponding positions are selected:

$$\mathbf{F}_i = H \cdot W_{\text{head}}^{i}, \quad \forall i \in [1, HW] \tag{28}$$

$$\mathbf{F}_{\text{pos}} = \left[\mathbf{F}[:, 1], \mathbf{F}[:, 2], \cdots, \mathbf{F}[:, \frac{H\times W}{32}]\right] \in \mathbb{R}^{(B\times HW)\times D} \tag{29}$$

where $W_{\text{head}}^{i} \in \mathbb{R}^{2C\times D}$ is the auxiliary transformation weight for the ith position.

Finally, the enhanced feature vector is obtained through score-weighted aggregation over the selected positions:

$$\mathbf{F}_{\text{weighted}} = \mathbf{F}_{\text{pos}} \odot M \in \mathbb{R}^{(B\times HW)\times D} \tag{30}$$

$$\mathbf{F}_{\text{out}} = \sum_{j=1}^{HW}\mathbf{F}_{\text{weighted}}^{(j)} \in \mathbb{R}^{B\times D} \tag{31}$$

where $\mathbf{F}_{\text{out}}$ is the final enhanced feature vector that can serve as input for subsequent recognition tasks. Since the retained scores participate in the aggregation, the gating parameters receive gradients from the classification loss, while the top-t selection itself is non-differentiable and only determines which positions are involved. Additionally, the gating scores s can be reshaped into a spatial activation map $A \in \mathbb{R}^{B\times H\times W}$ for visualizing the model's attention regions.

The output of MoEA is composed of the output Y of MoE and the output F_{out} of MoA concatenated along the channel dimension, effectively integrating the advantages of the two complementary features to form a more expressive mixed representation:

$$\mathbf{F}_{\text{MoEA}} = \text{Concat}(\mathbf{F}_{\text{out}}, Y) \in \mathbb{R}^{B\times 2D} \tag{32}$$

3.4. Loss function

In multi-label recognition tasks, each sample may simultaneously belong to multiple classes, and these classes are not mutually exclusive. The binary cross-entropy loss perfectly adapts to this multi-label characteristic by performing independent binary classification for each class. Furthermore, this loss function possesses favorable gradient properties mathematically, providing clear and stable optimization signals for each class, making it particularly suitable for handling the common class imbalance problem in sewer defect data. By independently processing each label, the model can more flexibly learn the complex relationships and co-occurrence patterns among different defect types.

Since the dataset is heavily imbalanced, we weight each positive observation of class c by the negative-to-positive observation ratio of that class. Let N denote the total number of images in the training split and N_c the number of images in the split containing at least one annotation of class c:

$$w_c = \frac{N - N_c}{N_c}, \tag{33}$$

where $N - N_c$ counts the images *not* containing class c (negative observations) and N_c the images containing it (positive observations). This way, the loss of minority classes is weighted higher when present in an image, while the loss of majority classes is weighted lower. The weighted binary cross-entropy loss is then computed as:

$$\mathcal{L}_{\text{W-BCE}} = -\frac{1}{B}\sum_{i=1}^{B}\sum_{c=1}^{NC}\Big[w_c\, Y_{i,c}\cdot\log\left(\sigma(\hat{Y}_{i,c})\right) + (1 - Y_{i,c})\cdot\log\left(1-\sigma(\hat{Y}_{i,c})\right)\Big], \tag{34}$$

where $\hat{Y} \in \mathbb{R}^{B\times NC}$ denotes the final logits of the model, $Y \in \{0,1\}^{B\times NC}$ denotes the ground-truth multi-label annotations, $\sigma(\cdot)$ is the Sigmoid function, and NC is the number of defect classes.

The total loss consists of the main task loss and the auxiliary regularization loss of the MoEA system:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{W-BCE}}(\hat{Y}, Y; w) + \mathcal{L}_{\text{aux}}. \tag{35}$$

4. Experiments and discussion

4.1. Dataset

Sewer-ML (Haurum & Moeslund, 2021) is a large-scale multi-label image dataset specifically designed for sewer defect recognition, comprising 1.3 million real-world images. It covers 17 common defect classes as shown in Table 3, which exhibit a pronounced long-tail distribution. A single image may contain up to eight different types of defects, highlighting the complex interdependencies among labels and the challenge of class imbalance in multi-label recognition. To reflect the practical significance of different defect types, the dataset also incorporates Class Importance Weights (CIW), indicating the criticality of each defect in actual maintenance scenarios. Since the test set is not publicly available, current studies primarily rely on its rigorously partitioned training and validation sets for experimental analysis. Furthermore, the dataset provides multimodal prior information in the form of sewer material.

Table 3
Defect codes, descriptions, and corresponding CIW values.

Code	Description	CIW
RB	Cracks, breaks, and collapses	1.0000
OB	Surface damage	0.5518
PF	Production error	0.2896
DE	Deformation	0.1622
FS	Displaced joint	0.6419
IS	Intruding sealing material	0.1847
RO	Roots	0.3559
IN	Infiltration	0.3131
AF	Settled deposits	0.0811
BE	Attached deposits	0.2275
FO	Obstacle	0.2477
GR	Branch pipe	0.0901
PH	Chiseled connection	0.4167
PB	Drilled connection	0.4167
OS	Lateral reinstatement cuts	0.9009
OP	Connection with transition profile	0.3829
OK	Connection with construction changes	0.4396

Table 4
Experimental environment configuration.

Parameter	Configuration
CPU	i7-14700KF
GPU	Nvidia GeForce RTX4090D(24 G)
Accelerated Environment	CUDA 12.2
Operating System	Ubuntu 22.04.3 LTS

XR-Sewer is a multi-label image dataset collected from sewers in Chengdu, Sichuan Province, China, containing 46,820 real inspection images gathered from 16 main urban districts. The dataset is split into training and test sets in an 8:2 ratio and includes 16 defect classes—one less "OK" defects than Sewer-ML. A distinctive characteristic of XR-Sewer is its rich multimodal metadata, including sewer material, the time of defect occurrence, and the city area where the defect was located.

To evaluate the model's generalization capability, we conducted experiments on the QV-Pipe dataset, which is a large-scale sewer inspection dataset comprising 9601 real-world pipeline videos. The dataset features 17 multi-label categories (16 defect types and one normal class) with a total duration exceeding 55 h. Characterized by long-tailed distribution and diverse real-world conditions, including various sewer materials and co-occurring defects, QV-Pipe serves as a valuable complement to the Sewer-ML dataset. Since the official test set is not publicly available, we divided the public training data into training and test subsets in a 7:3 ratio to ensure reproducible experimental evaluation.

4.2. Training procedure

To verify the effectiveness of the proposed method, we conduct neural network training and testing under the computer configuration parameters listed in Table 4. In the training phase, the learnable parameters of our method are trained from scratch using stochastic gradient descent (SGD) with a momentum of 0.9 and a weight decay of 1×10^{-4}. The total number of training epochs is 90, and the initial learning rate is 1×10^{-2}, scheduled by a cosine annealing strategy with a warm-up of 5 epochs. The batch size is set to 256, and the input images are scaled to 224×224. Data augmentation includes random flipping and random jittering of pixel values (brightness, contrast, saturation, and hue). The CLIP text encoder is used only during training to generate the material embeddings, which are precomputed and stored as fixed vectors for inference.

4.3. Evaluation metrics

For the multi-label sewer defect classification, false negatives often result in more serious consequences compared to false positives. Fol-

lowing Haurum and Moeslund (2021), we adopt $F2_{CIW}$ as the metrics to fully evaluate our method.

$$F2_{CIW} = \frac{\sum_{c=1}^{NC} F2_c \times CIW_c}{\sum_{c=1}^{NC} CIW_c} \tag{36}$$

where CIW_c and $F2_c$ are the CIW and F2 score for class c, respectively.

4.4. Comparison with SOTA methods

As shown in Table 5, among image multi-label classification backbone networks under 90M parameters, the Flash SewerNet-tiny model proposed in this paper (with only 12.11M parameters) achieved SOTA performance on the Sewer-ML dataset with a comprehensive $F2_{CIW}$ score of 64.65%, and achieved the best results on 8 out of 17 defect categories. What is particularly outstanding is that its parameter count is close to one-third of the second ranked PoolFormer-s36 model, but its performance is 2.52% higher, fully demonstrating its extremely high parameter efficiency and excellent recognition ability.

This superiority is further reflected in its performance alignment with defect category importance. Flash SewerNet-tiny achieves consistently strong results on the most critical defect types with the highest sample proportions: on FS, OB, and OK—the three most frequent categories in the dataset—it delivers highly competitive performance, and trailing the best-performing model by only 0.53, 0.88, and 0.53 percentage points, respectively, using less than half the parameters. More importantly, on challenging low-weight categories such as PB, IS, OP, and FO, it achieves 70.27%, 35.19%, 79.58%, and 25.06%, ranking first, second, first, and second among all compared methods, respectively. This consistent performance across the entire category spectrum demonstrates that Flash SewerNet-tiny effectively balances recognition capability across both dominant and rare defect patterns, making it particularly suitable for real-world sewer inspection scenarios where diverse defect types must be reliably identified.

To further evaluate the effectiveness of the proposed Flash SewerNet as a feature extractor, we integrate it into the SewerMan framework and compare with several state-of-the-art (SOTA) approaches that incorporate advanced techniques, as shown in Table 6. This comparison spans diverse backbone designs, including CNN-based architectures (Zhou, AutoSewerNet), traditional convolutional backbones (TResNet-L, ResNet variants), vision transformers (ViT-Base), and multimodal CLIP-based methods (Zhong-CLIP, Yan-CLIP), thereby positioning our approach comprehensively within the current literature.

With Flash SewerNet-tiny as the backbone, SewerMan achieves an $F2_{CIW}$ score of 66.45% ± 0.38 with only 15.57M parameters, surpassing all compared methods including the previous best multimodal Zhong-CLIP (65.06%, 108.1M). This represents a +1.39% improvement while using merely 14.4% of the parameters (15.57M vs. 108.1M), underscoring the representational efficiency of our lightweight backbone design. Notably, when equipped with Flash SewerNet-small, SewerMan further establishes a new SOTA on the Sewer-ML dataset with an $F2_{CIW}$ score of 67.24% ± 0.27, outperforming the previous best method by +2.18% while requiring less than half the parameters (48.46M vs. 108.1M). The small standard deviations (≤ 0.38) across ten independent runs confirm the robustness and reproducibility of our results. The consistent performance gains across both tiny and small variants, with substantially reduced parameter counts, confirm that the improvements stem from architectural effectiveness rather than model scaling alone. These results collectively demonstrate that Flash SewerNet serves as a highly effective feature extraction backbone, and when combined with the SewerMan architecture, achieves superior classification performance with significantly higher parameter efficiency for multi-label sewer defect classification.

These results primarily stem from two core capabilities of the Flash SewerNet architecture: first, its powerful ability to extract degraded visual features under sensor noise interference, effectively mitigating image quality issues caused by noise in enclosed sewer environments; second, its effective disambiguation of visual features through multimodal

Table 5
Comparison of F2 scores (in %) with different multi-label image classification backbones on the Sewer-ML validation set.

Method	Param	RB	OB	PF	DE	FS	IS	RO	IN	AF	BE	FO	GR	pH	PB	OS	OP	OK	$F2_{CIW}$
Xie (Yu et al., 2024)	9.17M	38.87	77.86	52.92	37.48	79.78	19.96	32.90	46.59	53.52	53.48	12.66	58.74	56.49	40.44	26.37	47.51	76.98	48.57
Hassan (Xie et al., 2019)	57.81M	0.00	52.60	7.32	7.37	65.87	3.30	10.29	0.00	27.24	0.00	0.00	21.86	11.94	0.00	0.00	0.00	0.00	13.14
Chen (Chen et al., 2018)	21.79M	38.82	66.38	31.00	20.81	73.90	12.77	43.33	33.70	42.02	45.98	11.64	43.15	52.73	19.18	20.63	51.75	66.21	42.03
Myrans (Hassan et al., 2019)	-	3.23	7.81	4.06	1.84	9.59	0.66	1.16	1.85	5.95	4.78	0.49	5.91	2.03	2.26	0.69	0.33	13.32	4.01
ResNet50 (He et al., 2016)	23.54M	38.46	80.60	51.84	37.25	84.54	22.73	35.16	45.67	54.91	54.51	16.36	59.11	57.30	48.58	35.30	57.46	79.36	51.97
ResNet101 (He et al., 2016)	42.53M	42.45	84.34	51.08	35.34	87.49	19.98	37.81	47.47	59.18	59.87	10.39	64.78	61.24	44.03	34.81	54.23	82.99	53.26
Efficientnet-b2 (Tan & Le, 2019)	7.66M	37.88	79.01	51.68	36.99	83.49	17.67	31.45	42.71	52.96	52.04	15.03	56.24	56.21	45.20	34.43	56.91	77.85	50.44
EfficientNet-b5 (Tan & Le, 2019)	28.20M	44.00	84.00	63.36	46.94	86.84	30.97	43.65	53.69	60.40	60.73	23.93	66.72	68.26	63.12	54.54	71.19	83.42	60.96
EfficientNet-v2-small (Tan & Le, 2021)	19.93M	38.51	81.09	59.01	43.13	84.89	20.36	37.70	48.21	55.33	55.14	15.62	59.14	63.76	59.08	45.34	65.96	79.66	55.60
ViT-base (Vaswani et al., 2017)	85.66M	37.48	80.20	55.57	39.99	81.42	22.19	34.00	46.67	54.11	53.80	14.35	57.46	57.80	52.37	35.32	58.13	77.41	51.74
Swin-Transformer-base (Liu et al., 2021)	86.70M	38.34	82.48	53.62	39.14	85.84	22.23	33.46	45.70	56.33	55.66	14.99	57.97	57.20	50.66	39.68	61.48	80.86	53.28
TResNet-m (Ridnik et al., 2021)	29.31M	43.55	84.49	51.02	37.23	87.79	21.75	37.57	47.93	60.01	59.99	10.51	65.61	64.43	43.62	33.70	57.04	83.71	53.83
TResNet-l (Ridnik et al., 2021)	53.50M	43.08	84.39	51.75	37.39	87.81	23.50	37.03	49.20	60.10	60.60	10.91	67.05	66.64	48.24	34.53	60.12	83.59	54.63
TResNet-xl (Ridnik et al., 2021)	75.69M	43.34	84.44	52.99	38.50	87.69	21.37	37.01	48.93	59.79	60.42	10.79	65.77	65.07	46.46	35.12	58.61	83.63	54.42
KSSNet (Wang et al., 2020)	47.52M	43.74	84.64	52.24	35.92	87.45	21.76	38.67	48.54	59.89	60.81	11.08	67.40	63.94	44.73	36.60	57.05	83.30	54.42
Benchmark (Haurum & Moeslund, 2021)	62.80M	42.71	82.48	48.33	34.06	85.73	24.28	43.69	46.66	57.81	60.37	9.84	65.08	64.05	44.74	32.82	60.23	82.83	53.49
R50-MLT (Haurum et al., 2022b)	23.50M	43.80	85.20	62.80	36.80	**88.10**	11.80	45.20	41.80	22.00	47.50	15.80	57.50	**75.80**	57.90	62.90	69.80	82.10	59.73
CT-GCN (Haurum et al., 2022b)	25.20M	43.00	84.10	64.90	37.50	87.20	9.80	49.80	46.90	23.50	51.50	11.00	59.80	74.90	66.80	64.80	73.20	82.90	61.35
CT-GAN (Haurum et al., 2022b)	24.20M	43.90	**85.90**	66.20	35.70	88.00	4.90	48.90	45.80	20.00	50.80	9.90	58.90	75.20	65.90	**65.10**	74.10	**83.80**	61.70
ConvNet-base (Liu et al., 2022c)	49.42M	44.68	83.60	63.93	46.34	87.42	35.19	50.18	50.68	61.14	61.51	25.53	66.61	68.14	61.77	52.12	72.21	83.38	61.15
PoolFormer-s36 (Yu et al., 2022)	30.30M	**45.23**	83.76	65.84	46.19	87.09	**37.11**	**50.80**	52.71	60.94	61.96	**30.57**	65.95	66.94	65.51	55.10	70.92	83.05	62.13
Mambaout-small (Yu & Wang, 2025)	48.50M	39.44	82.14	55.57	41.12	84.91	23.95	35.31	47.76	56.29	55.09	15.71	58.80	59.21	50.98	39.47	60.64	79.91	53.73
Flash SewerNet-tiny	12.11M	45.02	85.02	**69.58**	**52.20**	87.57	35.19	50.11	**56.05**	**61.25**	**62.61**	25.06	**68.26**	73.06	**70.27**	62.64	**79.58**	83.27	**64.65**

Table 6
Comparison of our network with SOTA methods on the Sewer-ML validation set. Results of Flash SewerNet are reported as mean ± std over ten independent runs.

Method	Backbone	Param (M)	$F2_{CIW}$ (%)
Hu (Hu et al., 2023)	TResNet-L	96.7	63.38
MSHViT (Haurum et al., 2022a)	ResNet50	29.42	61.68
MA-Q2L (Zuo et al., 2025)	ResNet101	143	63.64
Zhou (Zhou et al., 2023)	CNN-based	24.68	54.54
AutoSewerNet (Wang et al., 2025)	CNN-based	15.63	60.07
Zhong-CLIP (Zhong et al., 2025)	ViT-base	108.1	65.06
Yan-CLIP (Yan et al., 2024)	ViT-base	108.1	63.13
SewerMan (Zhong et al., 2026)	Flash SewerNet-tiny	**15.57**	66.45 ± 0.38
SewerMan (Zhong et al., 2026)	Flash SewerNet-small	48.46	**67.24 ± 0.27**

Table 7
Performance comparison on QV-Pipe dataset. Results of Flash SewerNet are reported as mean ± std over ten independent runs.

Method	mAP (%)	Macro-F1 (%)
SlowFast (Feichtenhofer et al., 2019)	44.89	54.68
TIN (Shao et al., 2020)	46.57	55.31
TANet (Hao et al., 2025)	48.23	31.94
TSM (Lin et al., 2019)	53.38	49.65
SVST (Liu et al., 2022a)	57.30	49.44
TMSDC (Zhao et al., 2023)	55.09	55.07
CSA (Rayhana et al., 2023)	54.32	55.19
TMFF (Hu et al., 2024)	60.66	**66.13**
POEN (Zhao et al., 2023)	62.69	64.14
Yan-CLIP (Yan et al., 2024)	63.35	64.26
Flash SewerNet-tiny	**63.36 ± 0.18**	65.87 ± 0.22

information fusion, leveraging textual prior information such as sewer material to assist in distinguishing visually similar yet categorically different defects, thereby significantly enhancing perceptual and decision-making reliability in complex scenarios. Notably, the model also excels in handling long-tail distributions, achieving top performance in rare defect classes with scarce samples, proving the effectiveness of the proposed approach in addressing class imbalance.

To further validate the generalization capability of Flash SewerNet, we conducted comprehensive evaluations on the challenging QV-Pipe dataset, which comprises ultra-short video clips captured from real-world sewer inspections. All compared methods, including ours and all baselines, are evaluated on the same 7:3 split of the public QV-Pipe training data. The video data are converted into images using a 3×3 supergraph-based frame sampling strategy, followed by video-level prediction aggregation. Based on the comprehensive evaluation for the sewer video recognition task in Table 7, our proposed Flash SewerNet-tiny demonstrates remarkable performance, achieving a mean Average Precision (mAP) of $63.36\% \pm 0.18$, which is comparable to the best-performing method Yan-CLIP (63.35% mAP). Notably, Flash SewerNet-tiny performs comparably to TMFF (60.66% mAP, 66.13% Macro-F1) in Macro-F1 ($65.87\% \pm 0.22$), despite being designed as a lightweight still-image model rather than a video-specific architecture. This is particularly significant given the ultra-short duration of QV-Pipe clips, where each clip contains only a few frames with limited temporal information, making frame-level feature quality the dominant factor for recognition performance. While TMFF with the Swin-B backbone attains the best Macro-F1 score (66.13%), this advantage comes with substantial computational overhead, as our architecture requires merely one-seventh of TMFF's parameter count. More importantly, the mAP margin over Yan-CLIP (0.01 percentage points) is far smaller than the standard deviation of our method (0.18), so we do not claim a statistically significant accuracy advantage; instead, the key advantage of Flash SewerNet-tiny lies in its parameter efficiency, achieving performance on par with Yan-CLIP (in the 108M parameter range) using only about 12M parameters. The small standard deviations (≤ 0.22) across ten independent runs confirm the robustness and reproducibility of our results on this challenging video benchmark. These results substantiate that our method achieves an optimal balance between model efficiency and recognition performance for sewer video analysis, offering a practical solution for real-world deployment where computational resources and operational efficiency are paramount.

These quantitative results fully validate that the Flash SewerNet model achieves the current best recognition accuracy in multi-label sewer defect recognition tasks with a compact and lightweight architecture, highlighting its dual advantages in both model efficiency and recognition accuracy.

4.5. Ablation studies

To validate the effectiveness of the proposed method, five ablation studies were conducted to evaluate the performance of the network, as summarized in Table 8.

The baseline model is primarily composed of residual blocks, with Gaussian–Laplacian kernel-enhanced edge information integration at the downsampling stage of each phase. It achieved an $F2_{CIW}$ score of 55.13% , a Micro-F1 score of 57.12% , a Macro-F1 score of 39.67% , and a 0–1 Exact Match score of 42.91% . The relatively low performance, particularly the significant gap between Macro-F1 and other metrics, reveals the model's weakness in recognizing tail classes under long-tailed distribution, mainly due to class imbalance in sewer defect data and feature degradation caused by sensor noise.

In Experiment II, the incorporation of the EA-Stem module further enhanced the model's edge perception capability. The Macro-F1 score relatively increased by 5.2% compared to the baseline. This improvement demonstrates the vital role of the LoG filter kernel in enhancing the model's ability to extract contour information of various defects: its second-order differential characteristics accurately locate regions with abrupt edge intensity changes, effectively strengthening the boundary feature expression of defects such as cracks and holes, thereby significantly improving recognition accuracy for morphologically diverse defects.

In Experiment III, the further addition of the GMMPFE module substantially enhanced the model's multi-label recognition capability for sewer defects. The $F2_{CIW}$ score relatively increased by 7.6% compared to Experiment II, with noticeable improvements in both head and tail classes. This is mainly attributed to the effective suppression of common noise interference from optical sensors in enclosed environments via multi-scale Gaussian filtering and the incorporation of sewer material text information as a supplement to visual features. A cross-modal attention mechanism helps disambiguate visually similar but categorically different defects, enhancing feature discriminability.

In Experiment IV, the MoEA module was introduced on the basis of Experiment II. The results showed a differentiated improvement in performance: the Micro-F1 score relatively increased by 1.9%, while the

Table 8
Ablation experimental results of Flash SewerNet-tiny.

Experiment	+ EA-Stem	+GMMPFE	+MoEA	$F2_{CIW}$(%)	Micro-F1(%)	Macro-F1(%)	0-1 Exact Match(%)
I				55.13	57.12	39.67	42.91
II	✓			56.97	60.14	41.96	45.05
III	✓	✓		61.33	64.33	46.24	48.48
IV	✓		✓	58.95	61.34	43.80	45.96
V	✓	✓	✓	**64.65**	**67.81**	**51.87**	**51.18**

Table 9
Ablation study on the number of experts in the MoEA module with $F2_{CIW}$(%) score on Sewer-ML dataset.

k/8	1/8	2/8	4/8	8/8
Flash SewerNet-tiny	63.60	**64.65**	64.19	62.86

Table 10
Performance comparison of baseline, MOE, and MoEA modules on Sewer-ML dataset.

Method	Param	FPS	$F2_{CIW}$(%)
Base + EA-Stem + GMMPFE	**4.33M**	**175**	61.33
Base + EA-Stem + GMMPFE + Transformer	14.03M	109	62.53
Base + EA-Stem + GMMPFE + MOE(2/8)	5.52M	119	62.92
Base + EA-Stem + GMMPFE + MoEA(2/8)	12.11M	104	**64.65**

Macro-F1 score relatively improved significantly by 4.3%. This discrepancy confirms that the MoEA module considerably enhances the recognition capability for tail classes in long-tailed data. The greater improvement in Macro-F1 indicates that, through expert specialization and assistant collaboration, the module effectively improves feature learning and classification accuracy for defect classes with scarce samples, thereby overall enhancing the model's ability to handle class imbalance.

In addition, we investigated the impact of the number of experts in the MoEA system. As shown in Table 9, with the total number of experts fixed at eight (aligning with the eight types of sewer materials), the $F2_{CIW}$ score first increased and then decreased as the number of experts grew: peaking at 64.65% with 2 experts and declining to 62.86% with 8 experts. This trend is not only due to overfitting risks and computational efficiency considerations but also because too many experts can fragment the consistency of the feature space and introduce routing ambiguity. The gating network is prone to decision bias under complex choices, which ultimately leads to performance degradation.

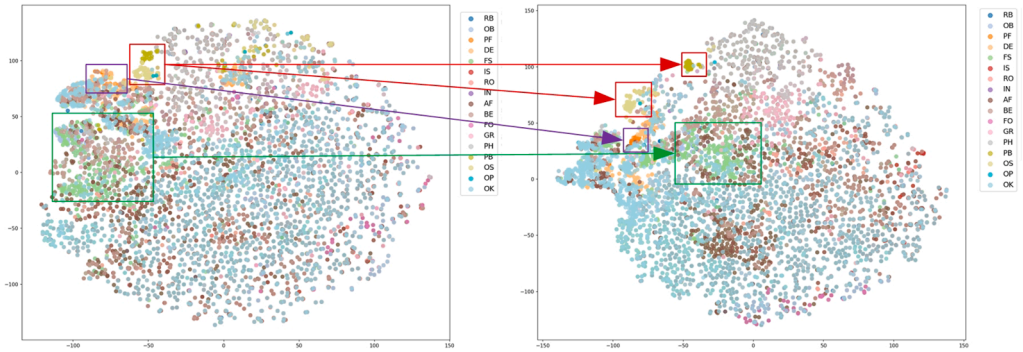
Additionally, compared to conventional MoE system, the MoEA module with assistant mechanisms leads to a substantial increase in parameters, mainly due to the additional parameters required for modeling fine-grained local features. However, the decrease in FPS is primarily influenced by two factors: the increased computational load from more experts and the more complex routing decisions required from the gating network. It is worth noting that the assistant system itself, with its parameter-sharing and parallel computation mechanisms, contributes relatively limited computational overhead alone, as shown in Table 10.

To isolate the architectural benefit of MoEA from increased capacity, we add a conventional Transformer-based classification head on top of Base + EA-Stem + GMMPFE. This model has 14.03M parameters, which is even slightly larger than the full MoEA model (12.11M). It achieves only 62.53% $F2_{CIW}$, while MoEA reaches 64.65% with fewer parameters (12.11M), yielding a +2.12 percentage point improvement. The Transformer head runs at 109 FPS versus 104 FPS for MoEA, indicating comparable inference speed. This confirms that the gain of MoEA does not come from increased capacity but from the assistant branch and the dual-constraint optimization.

In Experiment V, the integration of the three core modules on the baseline model yielded the SOTA result: an $F2_{CIW}$ score of 64.65% and a 0–1 Exact Match score of 51.18%. This fully demonstrates the multi-dimensional advantages of the synergy among the three modules: EA-Stem provides a structural perception foundation by enhancing edge feature extraction, GMMPFE improves feature discriminability through multimodal fusion and noise suppression, and MoEA optimizes decision-making under long-tailed distribution via expert specialization. Together, they form a hierarchical feature learning and decision-making framework that not only significantly improves overall recognition accuracy but also achieves a breakthrough in the strict exact match metric, indicating the model's ability to accurately predict multiple defect labels simultaneously and effectively address the challenge of collaborative recognition in complex sewer multi-label environments.

To further validate the effectiveness of our method in mitigating cross-category visual similarity, we conduct a feature-space visualization analysis using t-SNE. Specifically, we randomly sample 5% of the validation set and extract the penultimate-layer features from both the baseline Benchmark and our Flash SewerNet-tiny. To quantitatively assess clustering quality, we adopt the V-measure, defined as the harmonic mean of homogeneity and completeness, which provides a holistic evaluation of how well the learned features separate different defect classes.

As shown in Fig. 5, the baseline features exhibit severe inter-class overlap, with samples from different defect categories widely intermixed across the feature space. This issue is particularly pronounced for tail classes such as IS, PB, and PF, whose samples are scattered without forming coherent clusters—an observation that quantitatively corroborates the cross-similarity challenge identified in our introduction. In contrast, Flash SewerNet-tiny produces substantially more compact and well-separated clusters, characterized by clearly defined boundaries between most defect categories. Correspondingly, the V-measure score improves from 0.1057 to 0.1519 (ours), representing a relative improvement of 44%. This quantitative evidence, together with the visual inspection, confirms that the GMMPFE module's multi-scale Gaussian smoothing and multimodal semantic guidance effectively enhance feature discriminability, thereby mitigating the inter-class confusion that plagues existing methods.



(a) Benchmark
V-measure scores:0.1057

(a) Flash SewerNet-tiny
V-measure scores:0.1519

Fig. 5. t-SNE visualization of penultimate-layer features on a 5% random subset of the validation set. (a) Benchmark baseline: features exhibit severe inter-class overlap, with samples from different defect categories widely intermixed. (b) Flash SewerNet-tiny (ours): features form substantially more compact and well-separated clusters, demonstrating improved inter-class separability.

Table 11
Per-class F2 score (%) comparison on the Sewer-ML validation set. Classes are sorted by sample count (descending). Δ = Improvement over baseline.

Class	CIW	Benchmark (Haurum & Moeslund, 2021)	Flash SewerNet-tiny	Δ (%)
FS	0.6419	86.99	88.15	+1.16
OB	0.5518	83.56	85.97	+2.41
OK	0.4396	82.88	84.09	+1.21
AF	0.0811	59.64	61.29	+1.65
BE	0.2275	60.12	62.61	+2.49
GR	0.0901	66.39	67.39	+1.00
RB	1.0000	42.92	45.31	+2.39
IN	0.3131	50.41	55.77	+5.36
PH	0.4167	66.40	71.53	+5.13
RO	0.3559	37.17	50.72	+13.55
DE	0.1622	39.16	51.16	+12.00
PF	0.2896	54.06	69.38	+15.32
PB	0.4167	50.16	68.43	+18.27
IS	0.1847	23.89	34.63	+10.74
OP	0.3829	64.86	76.74	+11.88
FO	0.2477	11.30	25.18	+13.88
OS	0.9009	37.32	62.14	+24.82

4.6. Class-wise performance analysis

To rigorously evaluate the effectiveness of our proposed method on rare defect classes, we conduct a comprehensive class-wise performance analysis comparing Flash SewerNet-tiny against the Benchmark on the Sewer-ML validation set. This analysis focuses on two key aspects: (1) per-class F2 scores across all 17 defect categories, and (2) confusion matrices for the five rarest classes to reveal misclassification patterns

4.6.1. Per-class F2 score comparison

Table 11 presents the per-class F2 scores for all 17 defect categories, ordered by sample frequency from most common to rarest. The results demonstrate that Flash SewerNet-tiny achieves consistent and substantial improvements across tail seven classes, with an average gain of 15.76%, while preserving competitive performance on head classes (average +1.76%). The largest improvements are concentrated on the rarest defects: OS (+24.82%), PB (+18.27%), PF (+15.32%), FO (+13.88%), and DE (+12.00%), with IS improving from 23.89% to 34.63%, representing a relative improvement of 44.96% over the baseline. Furthermore, the improvements are skewed toward classes with higher CIW values: among five highly critical CIW classes (RB, OS, FS, OB, PB), our method achieves an average improvement of 9.81%, which is noticeably higher than the overall average of 8.43%. This alignment is practically significant, as improved performance on high-CIW classes directly translates to more reliable inspection outcomes for defects that matter most in real-world maintenance scenarios.

4.6.2. Confusion matrix analysis for tail classes

While per-class F2 scores provide a macroscopic view of performance, they do not reveal the nature of misclassifications—whether errors arise from false positives, false negatives, or confusion with specific competing classes. To gain deeper insight, we analyze the normalized confusion matrices for the five rarest defect classes, as shown in Fig. 6.

The accuracy improvements on these tail classes appear relatively modest in absolute terms—OS from 96.86% to 99.48%—which is expected given the extreme data scarcity: these classes account for fewer than 900 positive samples each, while the vast majority of images contain no such defects. However, the raw confusion counts reveal a more significant issue: the baseline exhibits a critical instability where false positives dramatically outnumber false negatives—for FO, FP/FN reaches 238:1 (22,629 vs. 95); for OS, 90:1 (4,036 vs. 45); for IS, 37:1 (8,998 vs. 242)—indicating that the baseline is overly sensitive, frequently misclassifying normal background regions as rare defects, which in real inspection would generate an overwhelming number of false alarms. In contrast, Flash SewerNet-tiny achieves substantial FP reductions: OS FP drops by 86.1% (4,036→560), FO by 91.7% (22,629→1,876), IS by 79.4% (8,998→1,852), and PB by 80.4% (4,134→811). This reduction is accompanied by a decrease in recall on these extremely rare classes, reflecting a precision–recall trade-off: the model shifts from an over-sensitive regime dominated by false alarms to a more balanced regime in which the remaining predictions are more reliable. These results confirm that the GMMPFE module's multi-scale Gaussian smoothing reduces the model's tendency to mistake background textures for tail-class signals, while the MoEA assistant mechanism prevents the systematic over-prediction observed in the baseline.

4.7. Analyzing the effect of sewer materials on multi-label sewer defect classification on the sewer-ML validation set.

To quantitatively assess the sensitivity of our model to the quality and correctness of multimodal priors, we conduct counterfactual experiments by systematically manipulating the text input to the GMMPFE module. Specifically, we compare six configurations: (1) Correct Material: using the actual sewer material text as the multimodal prior; (2) Text Ablation: removing the text input entirely, with the attention module operating solely on visual statistics; (3) Random Text: replacing the correct material description with random strings, providing semantically meaningless guidance; and (4) Mismatched Text: deliberately assigning an incorrect material label. (5) Learned Embedding: replacing the CLIP text embedding with a learnable embedding table indexed by the material ID, which provides the same material category information but contains no semantic text prior; and (6) One-hot Encoding: representing the material category as a one-hot vector followed by a learnable linear projection to the same dimension, again providing only categorical information without semantic content. The results are summarized in Table 12.

Replacing correct material text with random strings causes a dramatic performance drop to 47.74% $F2_{CIW}$—far below the text-ablated baseline (62.68%)—indicating that uninformative text actively corrupts feature selection and confirming that the network relies on meaning-

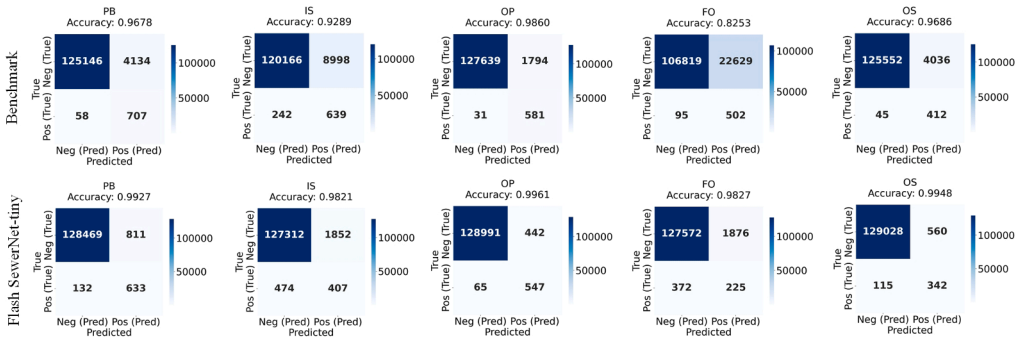


Fig. 6. Confusion matrix statistics for the five rarest classes on the Sewer-ML validation set.

Table 12
Counterfactual analysis on sewer materials fusion.

Text Input Configuration	$F2_{CIW}$(%)
Random Text	47.74
Mismatched Material Text	60.82
Text Ablated	62.68
One-hot Encoding	62.82
Learned-Embedding	63.30
Correct Material Text	**64.65**

Table 13
Performance of multimodal fusion strategies on XR-Sewer dataset.

Method	$F2_{CIW}$(%)
Flash SewerNet-tiny (Data)	51.45
Flash SewerNet-tiny (Area)	51.92
Flash SewerNet-tiny (Material)	52.57
Flash SewerNet-tiny + WA	53.03
Flash SewerNet-tiny + MLP	53.27
Flash SewerNet-tiny + AM	**54.45**

ful semantic alignment rather than mere architectural augmentation. When text is deliberately mismatched to the actual sewer material, performance falls to 60.82%, even lower than the text-ablated variant; this material-specific penalty demonstrates that the attention mechanism is actively guided by material semantics—incorrect guidance leads to worse confusion than no guidance. Finally, comparing correct text (64.65%) with text ablation (62.68%) isolates a +1.97 percentage point gain attributable purely to semantic priors after accounting for the effect of additional parameters. Collectively, these results demonstrate that the model is highly sensitive to the correctness and semantic quality of multimodal priors, and incorrect or noisy priors can actively harm performance—underscoring the importance of reliable material annotations in real-world deployment. To further examine whether the material prior requires CLIP-derived semantic guidance, we replace the CLIP text embedding with two simpler material representations while keeping the attention-based fusion mechanism and the parameter budget unchanged: a one-hot encoding of the material ID followed by a learnable linear projection, and a learned categorical embedding for each material type. The one-hot encoding achieves 62.82% $F2_{CIW}$, only marginally above the text-ablated baseline (62.68%) and 1.83 percentage points below the CLIP-based variant (64.65%), indicating that supplying the material category alone provides almost no benefit beyond the ablated model. The learned categorical embedding achieves 63.30% $F2_{CIW}$, which is 1.35 percentage points lower than the CLIP-based variant (64.65%). More importantly, the performance gap is mainly concentrated on tail classes: because sewer defect data exhibit a pronounced long-tailed distribution, the learned categorical embedding receives insufficient supervision for rare material–defect combinations and thus fails to form discriminative representations for these cases. In contrast, the CLIP-derived semantic embedding provides a structured and transferable prior that remains informative even when material–defect co-occurrences are scarce. Together, the monotonic ordering from one-hot encoding (62.82%) to learned embedding (63.30%) to CLIP-based embedding (64.65%) confirms that the benefit of the material prior does not come merely from supplying material identity as an additional input, but from the semantically structured CLIP embedding space, which is essential for guiding feature selection under long-tailed distributions.

4.8. Systematic validation of multimodal text information for enhancing multi-label sewer defect recognition

To comprehensively evaluate the effectiveness of multimodal textual information in improving multi-label sewer defect recognition, this study conducted an in-depth comparative experiment on the XR-Sewer dataset. Based on the Flash SewerNet-tiny model pre-trained on the Sewer-ML dataset, a 30-epoch fine-tuning process was performed. We first evaluated the individual contribution of each single-modal prior, then systematically explored multimodal fusion strategies. As shown in Table 13, four input configurations were designed: a baseline model using only image data, and comparative setups incorporating area, material, and time information. These configurations were evaluated using three fusion strategies—Weighted Addition (WA), Multilayer Perceptron (MLP) fusion, and Attention Mechanism (AM)—to thoroughly assess the impact of both the richness of multimodal information and the choice of fusion strategy on cross-dataset performance, as illustrated in Fig. 7.

The experimental results reveal several important findings. First, among individual priors, Material provides the largest improvement, followed by Area, confirming that sewer material is the most informative semantic prior for defect recognition. The introduction of additional multimodal information led to consistent improvements across all fusion strategies: Weighted Addition increased the $F2_{CIW}$ score to 53.03% (+0.46%), MLP fusion further improved it to 53.27% (+0.70%), and the AM achieved the highest performance with a score of 54.45% (+1.88%), establishing it as the most effective fusion method.

These results indicate that richer multimodal textual information provides more comprehensive contextual priors, significantly enhancing the model's ability to discriminate visually ambiguous features. The AM, with its dynamic feature alignment and adaptive weighting capabilities, most effectively leverages textual modal information. The consistent performance improvement with both increased modal richness and finer fusion strategies underscores the importance of multimodal learning in multi-label sewer defect recognition tasks.

4.9. Visualization and analysis

To further analyze the recognition capability of the Flash SewerNet model in real sewer environments, this study conducted multiple sets of visualization experiments. As shown in Fig. 8, a complex sewer image containing multiple defects—with ground truth labels including RO, FS, OB, and RB—was used for evaluation. Flash SewerNet successfully predicted all defect classes, and its Class Activation Mapping (CAM) visualizations exhibited highly concentrated response regions, accurately highlighting the defective structures. In contrast, Convnet-base only detected three defects and produced scattered heatmap responses, indicating its limited ability to localize critical features.

Fig. 9 further compares the model performance in the presence of annotation noise. Although the ground truth annotations for this image included only three defect classes—with the FS defect missed due to its location in a low-light area—Flash SewerNet not only correctly

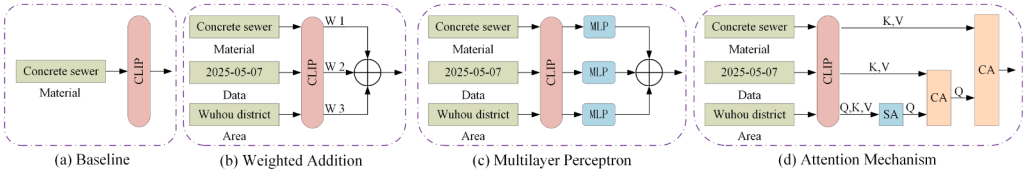


Fig. 7. Comparison of multimodal fusion strategies under different input configurations. SA and CA refer to Self-Attention and Cross-Attention, respectively.

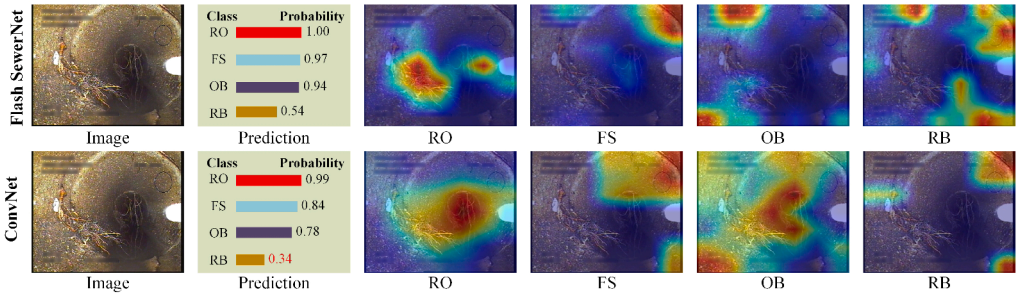


Fig. 8. Model visualization.The two rows represent the visualization results of Flash SewerNet and ConvNet, respectively. The heatmaps utilize a color gradient ranging from blue to red, where blue indicates low attention and red represents high attention.

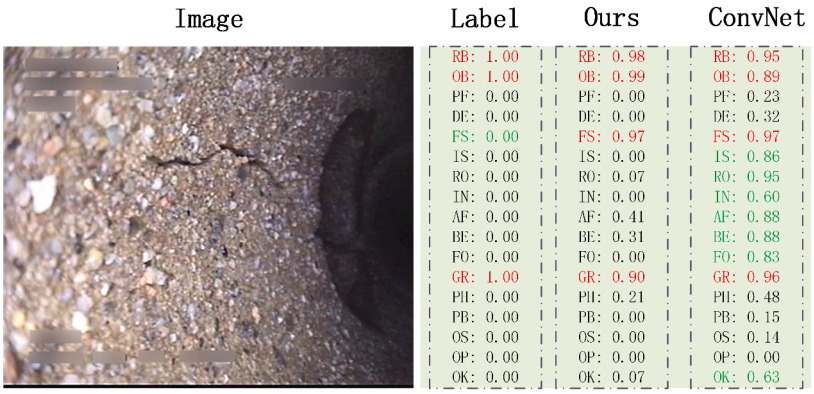


Fig. 9. Superior feature extraction enables robust defect recognition despite annotation noise and low-light conditions.

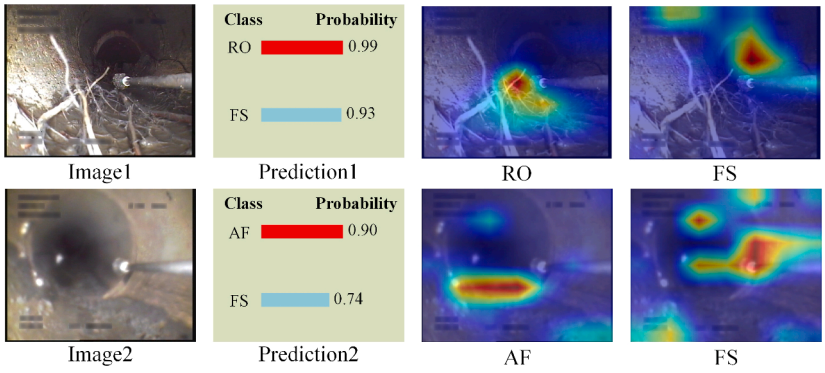


Fig. 10. Robust performance of Flash SewerNet under significant sensor noise.

identified all annotated classes but also accurately detected the unannotated FS defect. This demonstrates the model's high sensitivity to subtle features and strong robustness. In comparison, while Convnet-base correctly predicted some defects, it also produced seven false positives, suggesting its difficulty in distinguishing actual defects from background interference. This observation is consistent with the dispersed activation patterns observed in Fig. 8.

Subsequently, Fig. 10 illustrates the recognition performance of Flash SewerNet under significant sensor noise. Even in these low-quality, high-noise conditions, Flash SewerNet maintained high prediction accuracy, with its heatmaps consistently focusing on semantically salient defect regions. This indicates that the model effectively suppresses noise interference through Gaussian modeling and multimodal fusion, thereby enhancing the stability of feature representation.

While these visualizations are qualitatively encouraging, they alone cannot establish robustness to sensor degradation. To quantitatively evaluate robustness to sensor degradation, we conduct a controlled noise experiment on the Sewer-ML validation set. We randomly add noise to the input images at three severity levels (0.2, 0.4, and 0.6), covering Gaussian noise, salt-and-pepper noise, and blur. The noise parameters are set as `noise_std = 0.05`, `noise_amount = 0.02`, `blur_radius = 1.5`, and `blur_kernel = 5`. We compare the $F2_{CIW}$ scores of the baseline and Flash SewerNet-tiny, and report the *retention rate* relative to clean performance (The clean $F2_{CIW}$ is 55.13% for the baseline and 64.65% for Flash SewerNet-tiny.) And the results are summarized in Table 14.

As shown in Table 14, both models degrade as noise severity increases, but Flash SewerNet-tiny consistently retains a higher proportion of its clean performance. At severity 0.2, the retention rates are 96.98% (ours) vs. 93.92% (baseline); at 0.4, 94.17% vs. 89.24%; at 0.6, 90.92% vs. 85.54%. The retention gap widens from +3.06 to +5.38 percentage points as noise becomes more severe, indicating slower degradation and stronger noise robustness. These results provide controlled quantitative evidence that the proposed Gaussian-based noise suppression and multi-

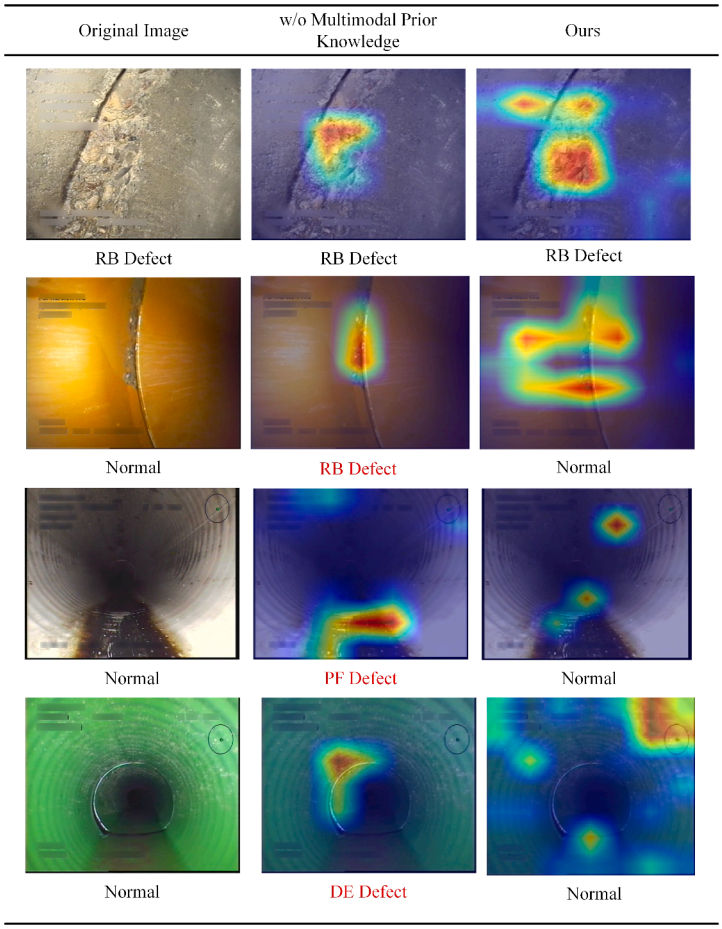


Fig. 11. Attention map comparison with and without material priors.

Table 14
Controlled noise robustness comparison on the Sewer-ML validation set. Values are $F2_{CIW}$ (%), with retention rate (%) relative to clean performance in parentheses.

Noise Severity	Baseline	Flash SewerNet-tiny
0.2	51.78 (93.92)	62.70 (96.98)
0.4	49.20 (89.24)	60.88 (94.17)
0.6	47.16 (85.54)	58.78 (90.92)

modal fusion improve robustness under sensor degradation, rather than merely producing visually plausible attention maps.

Finally, Fig. 11 illustrates how multimodal material priors influence feature selection and decision-making, we visualize the spatial attention maps generated by the GMMPFE module with and without material priors. In the first row, a concrete sewer with actual RB is correctly identified by both configurations, as the visual defect features are sufficiently salient. The second row presents a lined sewer in normal condition but with attached debris on its surface. Without material prior, the model incorrectly activates on the debris region and misclassifies it as RB, since the visual texture closely resembles the crack pattern in the first row. With the correct material prior, however, the model suppresses this false activation and correctly recognizes the region as normal background. The third row shows a PVC sewer with black dye staining at the bottom. Without material prior, the model misinterprets the staining as PF due to its visual similarity to sediment accumulation. With the material prior, the attention map shifts away from the staining and focuses on actual sewer surface features, correctly avoiding false positives. The fourth row depicts a corrugated PVC sewer with periodic structural patterns. Without material prior, the model incorrectly activates on the regular corrugation texture and misclassifies it as DE. With the material prior, the model recognizes the periodic pattern as a normal structural feature rather than a defect, suppressing the false activation.

In summary, the visualization experiments demonstrate that Flash SewerNet not only achieves higher defect recognition accuracy but also produces more concentrated and interpretable heatmap responses. Critically, these visualizations reveal that with material priors, the model does not rely solely on visual features—it actively incorporates material context to determine whether a visually salient region constitutes an actual defect or a normal material-specific artifact, effectively distinguishing defect from non-defect patterns that are visually similar but semantically distinct. This mechanism substantially reduces false positives in real-world sewer inspection scenarios. The model also exhibits strong robustness to annotation omissions and sensor noise, confirming its ability to adapt to complex and variable sewer imaging environments and further validating its effectiveness in practical applications.

4.10. Complexity analysis

To comprehensively evaluate the computational efficiency of our proposed method, we analyze four key metrics: parameter count (Params), theoretical floating-point operations (FLOPs), inference latency, and GPU memory consumption. For edge deployment evaluation, we additionally report inference speed on a Jetson Orin NX module (8-core ARM processor, 1024 CUDA cores, providing up to 100 TOPS of AI computing power).

As shown in Table 15, Flash SewerNet-tiny achieves the highest $F2_{CIW}$ score (64.65%) among all compared methods, outperforming LWEGNet by +6.57% and EfficientNet-v2-small by +8.05%. While

Table 15
Complexity comparison across lightweight architectures.

Method	Params (M)	FLOPs (G)	Memory (MB)	FPS	$F2_{CIW}$(%)
MobileNetV3 (Howard et al., 2019)	5.48	**0.06**	**17.70**	**390.15**	50.14
EfficientNet-b2 (Tan & Le, 2019)	7.66	0.66	89.00	22.36	50.44
LWEGNet (Lu et al., 2025)	**3.66**	1.47	44.27	171.76	58.08
EfffcientNet-v2-small (Tan & Le, 2021)	19.93	2.71	232.75	98.63	56.60
Flash SewerNet-tiny (w/o MoEA)	4.33	1.57	50.12	175	61.33
Flash SewerNet-tiny	12.11	1.80	157.35	104	**64.65**

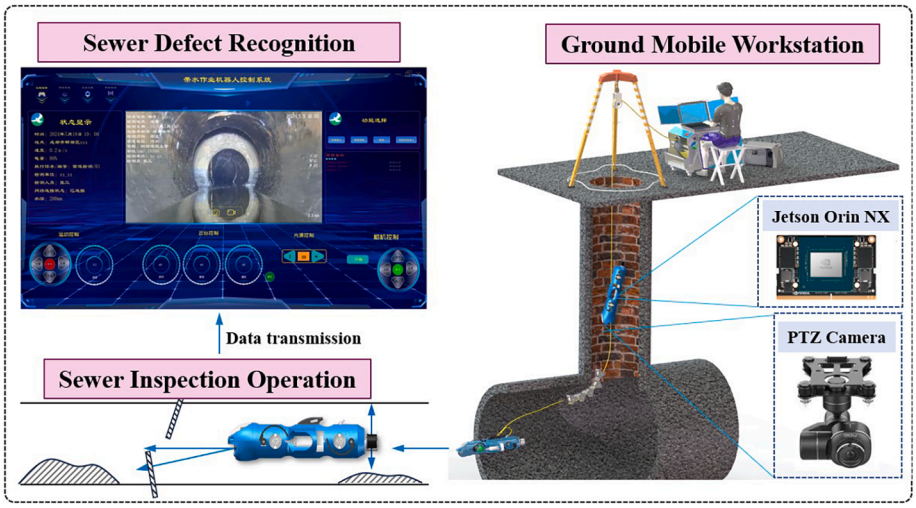


Fig. 12. The sewer inspection system.

MobileNetV3 exhibits superior efficiency in terms of FLOPs (0.06G), memory (17.70 MB), and FPS (390.15), its recognition performance (50.14%) lags considerably behind ours by 14.51 percentage points. This performance gap is not surprising, as generic lightweight backbones with simple architectures lack the capacity to capture the intricate visual patterns and complex dependencies inherent in sewer defect recognition—where defects exhibit high morphological diversity, severe sensor noise, and substantial cross-material interference. By contrast, Flash SewerNet is specifically tailored to address these challenging conditions through its multi-scale Gaussian smoothing and multimodal semantic guidance, enabling superior accuracy while maintaining practical efficiency. Importantly, Flash SewerNet-tiny requires only 157.35 MB of GPU memory and achieves 104 FPS on an RTX 4090d—well within the capabilities of modern edge devices.

It is also worth noting that the CLIP text encoder is used exclusively during training to generate the eight material-type embedding vectors corresponding to the sewer material categories. During inference, these embeddings are precomputed and stored as fixed vectors, bypassing the CLIP text encoder entirely and eliminating its computational cost. This design choice makes Flash SewerNet substantially more efficient than conventional vision-language models that require joint image-text encoding at test time, further reducing the practical inference burden.

To verify the practical effectiveness of the proposed method on edge devices, we have constructed a complete underground sewer defect recognition system, as shown in Fig. 12. The system consists of three parts: a ground mobile workstation, an inspection robot, and a remote management platform. The inspection robot is equipped with a 360° Pan-Tilt-Zoom camera for real-time acquisition of images inside the sewer, and is equipped with a Jetson Orin NX module for embedded inference. In practical environments, the inference speed of Flash SewerNet tiny on this device reaches 28.18 FPS, fully meeting the real-time requirements of edge devices in field sewer inspections.

Beyond raw inference speed, practical edge deployment entails additional constraints. In our specific scenario, the inspection robot is connected via a tethered cable that supplies both power and data transmission (Fig. 12), rendering battery life a non-issue and allowing the Jetson Orin NX to operate at its maximum 15 W mode. Regarding latency variation, our MoEA employs deterministic top-k routing with fixed computational paths, ensuring predictable per-frame inference time without stochastic branches. The model consumes less than 200 MB of the 8 GB unified memory, leaving ample capacity for system processes. Finally, the fully deterministic inference pipeline—comprising fixed convolution kernels, precomputed CLIP embeddings, and rule-based expert routing without stateful components—guarantees consistent outputs over extended operation, preventing cumulative error or memory leakage. In practical field tests lasting several hours, we observed no performance degradation or system instability.

4.11. Limitations and future work

Despite the promising results, our framework has two main limitations. First, it relies on material annotations to generate multimodal priors. As shown in our counterfactual experiments, incorrect or mismatched text guidance actively degrades performance, indicating that the model depends on meaningful semantic alignment. In real-world deployments, such metadata may be incomplete or unavailable, particularly for aging sewer systems with poor documentation. Second, the current framework operates on individual still images, treating each frame independently without exploiting the temporal continuity inherently present in sewer inspection videos. This limitation is particularly significant given the nature of real-world inspection data: typical CCTV inspection campaigns produce long, continuous videos spanning several hours and covering kilometers of pipeline, where defects evolve gradually across hundreds or thousands of consecutive frames. In contrast, existing benchmarks such as QV-Pipe dataset, which consists of short, isolated video clips with limited temporal context—each clip captures only a brief segment of a defect, without the extended before-and-after footage that would enable temporal reasoning. By not modeling these

long-range temporal dependencies present in full inspection videos, our approach may miss critical opportunities to: (1) reinforce predictions through frame-level consensus across extended sequences, (2) track defect progression and severity changes over time, and (3) distinguish transient artifacts from persistent structural defects that consistently appear across many frames.

Beyond these limitations, it is worth clarifying which components of Flash SewerNet are specific to sewer inspection and which are more broadly applicable to other industrial vision tasks, since noise, visually similar classes, imbalanced data, and limited computational resources are common across many industrial inspection problems. The EA-Stem module is largely task-agnostic: by combining Gaussian smoothing with the Laplacian operator, it targets low-level feature degradation caused by sensor noise and illumination variation, which is common to many industrial inspection scenarios, and it relies on no sewer-specific assumption. The GMMPFE module contains both general and task-specific elements: its multi-scale Gaussian smoothing and channel-attention fusion are general, whereas the use of sewer material as a semantic prior is domain-specific. The underlying principle—injecting a domain attribute, such as material or surface type, as a semantic prior to disambiguate visually similar classes—can be transferred to other industrial inspection tasks where such attributes are available. The MoEA decision system is also general, since it addresses long-tailed class distributions and is not tied to sewer defects. In contrast, the material taxonomy and the CLIP text prompts used in this work are sewer-specific and would need to be redefined for other domains.

To address these limitations, future work will focus on three directions: (1) We plan to explore self-supervised or weakly-supervised strategies that can infer material or environmental priors directly from visual patterns, reducing the dependency on manual metadata. (2) We will extend Flash SewerNet to incorporate long-range sequential dependencies across frames through lightweight temporal modules that preserve real-time performance on edge devices. Unlike prior video understanding methods designed for short clips, these modules must efficiently handle the unique challenges of hour-long inspection videos, including gradual lighting variations, periodic viewpoint jitter, and the need to maintain temporal coherence over thousands of frames without prohibitive memory or computational cost. This extension would enable improved robustness through temporal consensus and continuous defect tracking throughout the entire inspection sequence. (3) We plan to investigate the integration of prior knowledge from fluid dynamics models (Abbas et al., 2023; Nazar et al., 2026; Ramzan et al., 2025) to enhance the understanding of defect evolution under varying flow conditions, thereby improving the predictive capability and physical consistency of the recognition system. By linking observed visual patterns to the underlying physical processes that govern defect initiation and propagation, this physics-informed approach could provide a principled basis for distinguishing genuine structural deterioration from superficial anomalies.

5. Conclusion

To address three core challenges in multi-label sewer defect recognition—severe image degradation due to sensor noise, high visual similarity among defect classes and sewer materials, and recognition bias caused by long-tailed class distribution—this paper introduces Flash SewerNet, a lightweight expert network. The proposed method incorporates three specialized components: an EA-Stem module to enhance edge and contour feature extraction; a GMMPFE module that performs noise modeling and integrates multimodal textual information to improve feature robustness under noisy conditions and alleviate visual ambiguity; and an MoEA hybrid expert system that optimizes feature learning and classification under long-tailed data distribution. Extensive experiments on the Sewer-ML dataset demonstrate that Flash SewerNet-tiny achieves SOTA performance with an $F2_{CIW}$ score of 64.65% using only 12.11M parameters. The model also exhibits robust generalization on the QV-Pipe video benchmark (63.36% mAP, 65.87% Macro-F1) and achieves 28.2 FPS on the Jetson Orin NX edge device, confirming its practical deployability for real-time sewer inspection.

Despite these achievements, our framework has two main limitations. First, it relies on accurate material annotations for multimodal priors. Second, it operates on still images without exploiting temporal cues in video sequences.

Future work will focus on three directions: (1) video-based temporal modeling using recurrent or transformer architectures to capture cross-frame dependencies; (2) domain adaptation techniques to generalize across diverse sewer environments, pipe materials, and inspection conditions; and (3) integrating physical priors from fluid models to enhance the understanding of defect evolution under varying flow conditions, thereby improving predictive capability and physical consistency.

CRediT authorship contribution statement

Songyi Dian: Writing – review & editing, Supervision, Project administration, Investigation; **Xuke Zhong:** Writing – original draft, Visualization, Resources, Methodology, Data curation; **Yuzhong Zhong:** Methodology, Formal analysis, Conceptualization; **Xingxing You:** Validation, Software, Methodology; **Yafeng Zou:** Software, Formal analysis; **Quan Xiao:** Supervision, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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